

Neural DTS: Integrating Hyperbolic Classifiers into Natural Language Inference Systems

Honoka Kobayashi, Hinari Daido, Daisuke Bekki

Ochanomizu University

kobayashi.honoka@is.ocha.ac.jp

NALOMA VI 4 Aug (Tue)

The task: Natural Language Inference (NLI)

Natural Language Inference (NLI) is the task of determining the relationship between given **premises** and a **hypothesis**

└ Yes / No / Unknown

Examples of NLI:

P : A dog **and** a cat run.

H : A dog runs.



Yes

P : A dog **or** a cat run.

H : A dog runs.



Unknown

⇒ Reasoning over **meaning** — including **knowledge** not written in the premises

Two paradigms for answering it

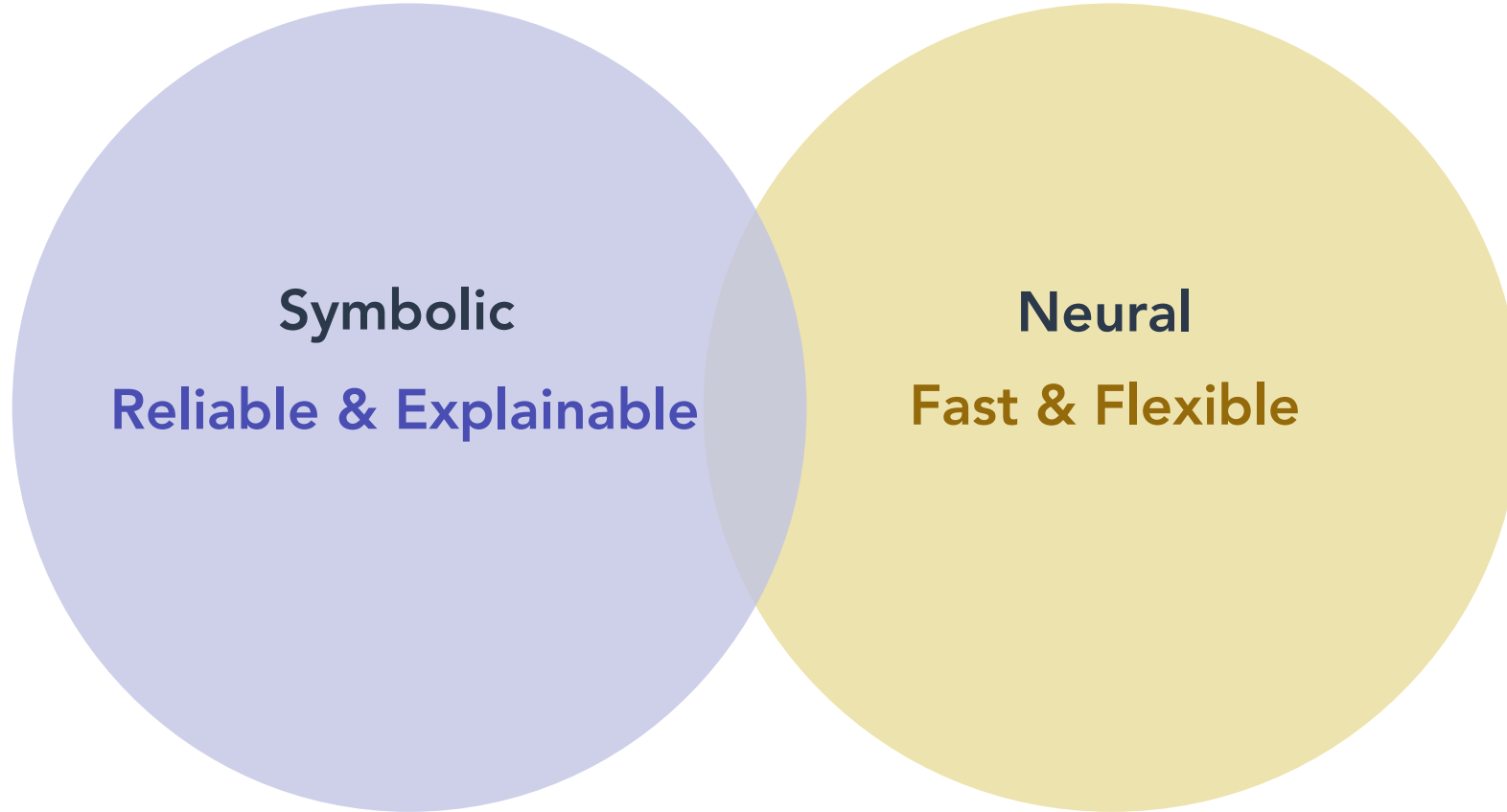
Symbolic / logical

- Translate each sentence into a meaning representation
 - Prove the hypothesis
 - if a proof is found, the answer is guaranteed correct
 - transparent: every rule is explicit
- world knowledge must be written by hand

Neural / data-driven

- Learns broad knowledge directly from data
 - Stays robust to changes in wording
 - Needs no hand-written rules
- Reasoning is a black box
- No guarantee of logical consistency

The tension, at a glance



Keep the **symbolic guarantees**, borrow the **neural ability to learn**

What does an NLI benchmark actually measure?

Score alone

End-to-end models reach high scores.
But evidence of what?

↔ nothing specifies what counts as
evidence of **understanding**

Our commitment

Meaning = proof conditions

↔ understanding
= mastery of **inferential role**

Only a theory of meaning can tell what a benchmark measures
DTS fixes what counts as a proof ⇒ NLI measures understanding

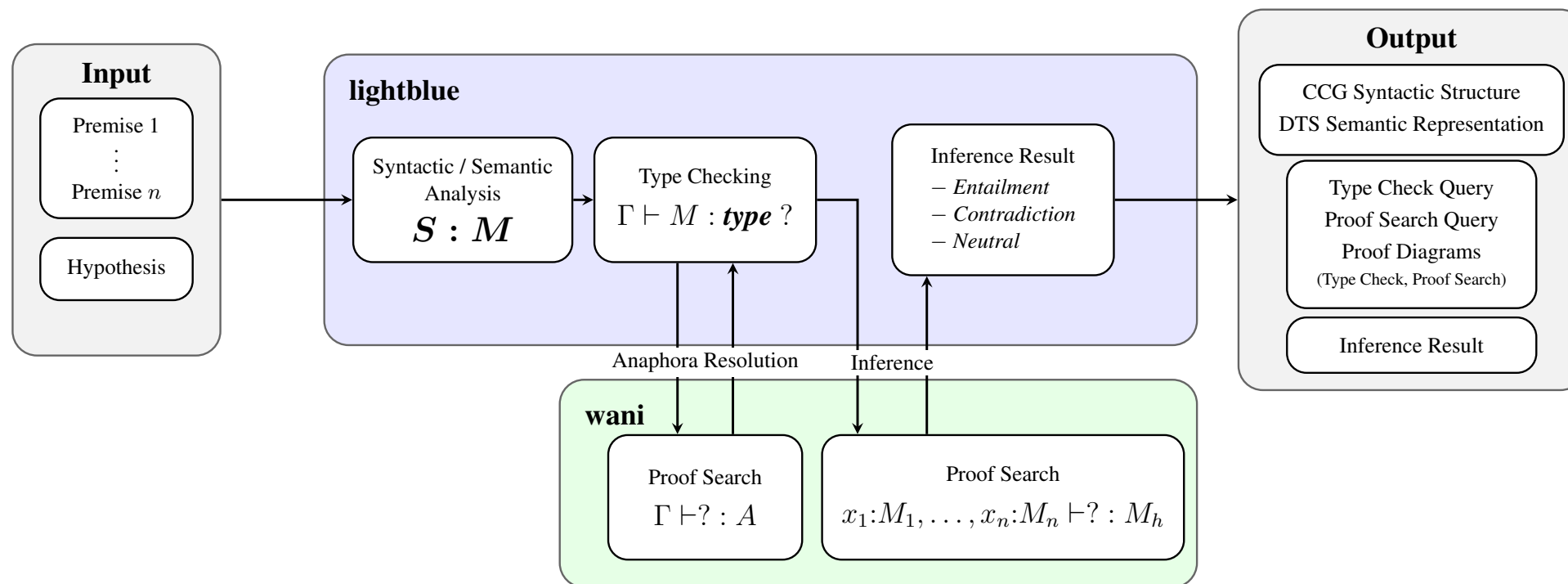
Dependent Type Semantics (DTS)

- A proof-theoretic semantics on Dependent Type Theory [Martin-Löf 1984].
- " P entails H " becomes "there is a proof of H from P ".
- The derivation is an explicit proof diagram $\Rightarrow$ explainable inference.

$$\begin{array}{c}
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash s_0 : \left[\begin{array}{l} u_3 : \left[\begin{array}{l} x_3 : \text{entity} \\ x_4 : \text{entity} \\ \text{apple}(x_4, x_3) \end{array} \right] \\ x_5 : \text{entity} \\ \text{eat}(x_5, \text{Taro}, \pi_1(u_3)) \end{array} \right]} \quad (\text{Var}) \\
 \frac{}{\quad} \quad (\Sigma E) \\
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash \pi_1(\pi_2(\pi_2(\pi_1(s_0)))) : \text{apple}(\pi_1(\pi_2(\pi_1(s_0))), \pi_1(\pi_1(s_0)))} \quad (\text{Con}) \\
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash \text{oracle}(\pi_1(\pi_2(\pi_2(\pi_1(s_0)))) : \text{fruit}(\pi_1(\pi_2(\pi_1(s_0))), \pi_1(\pi_1(s_0)))} \quad (\text{Con})
 \end{array}$$

Our symbolic backbone: the NLI pipeline [Tomita+ 2025]

Built from **lightblue** [Bekki & Kawazoe 2016] + **wani** [Daido & Bekki 2020]



Where the symbolic side breaks: lexical knowledge

The system needs **world knowledge**, which must be given **by hand** as axioms

P : A swallow crosses the sea.

H : A bird crosses the sea.



needs : A swallow is a bird

P : All noodles are cheap.

H : Udon is cheap.



needs : Udon is a noodle

- ▶ Cannot write all such facts in advance
- ▶ Picking only the few axioms we need, out of millions = **the flame problem**

Choosing the relevant fact from a huge space is exactly
what **neural methods** are good at

Our idea: Neural DTS [Bekki+ 2021, 2022] [linuma & Bekki 2025]

Keep the prover; let a neural component supply the missing axioms on the fly.

Previous work :

- Accuracy fell short — MLP F1 ≈ 0.77 , NTN ≈ 0.50 (near-random)
- Root cause: sparse *entity* distribution (predicates augmented, entities not)
- Not yet integrated into the prover — classifier tested stand-alone

This work :

- Narrow down where to put the neural part
- Judging word inclusion $\Rightarrow$ **Hyperbolic Classifier**

Prover = language faculty Classifier = general cognition
e.g. does swallow $\subset$ bird holds ?

The neural interface: Oracle

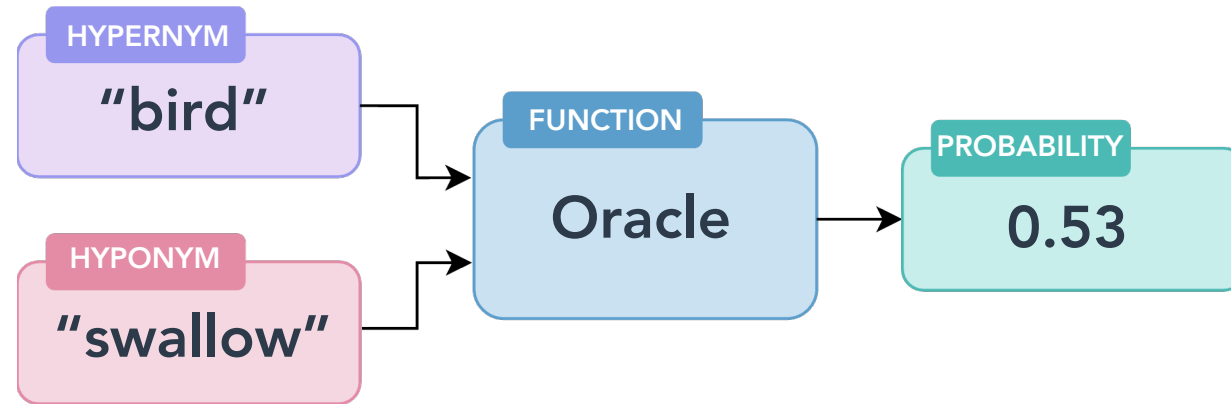
Goal: supply the missing world knowledge **automatically** and **probablistically**

P : A swallow crosses the sea.

H : A bird crosses the sea.



needs : A swallow is a bird



Implemented in **Haskell**, connected this to the pipeline:

Oracle :: DTTdB.ConName -> DTTdB.ConName -> Float

Which embedding? Comparison of methods

Embedding	Structure	Space
Word2Vec [Mikolov et al., 2013]	inner product	Euclidean
Order Embedding [Vendov et al., 2016]	order	Euclidean
Poincare Embedding [Nickel and Kiela, 2017]	distance	Hyperbolic
Hyperbolic Entailment Cones [Ganea et al., 2018]	order	Hyperbolic

The only method that combines **order** and **hyperbolic space** ➡ **Hyperbolic Entailment Cones**

Why hyperbolic Cones?

A hypernym-hyponym tree grows **exponentially**, as deeper levels have many more words

- **Euclidean** : needs high dimension (hundreds+)
- **Hyperbolic** : expands exponentially (≤ 10 dims)

Each word = **cone**

more specific word sits inside the cone of a general one
This asymmetric design matches the "is-a" relation

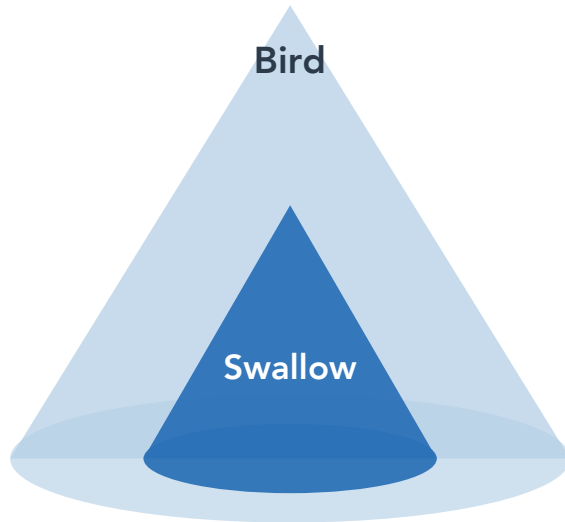


[Escher, Circle Limit III]

Oracle geometry: a cone is an "is-a" detector

Every word is a **cone** in the disk

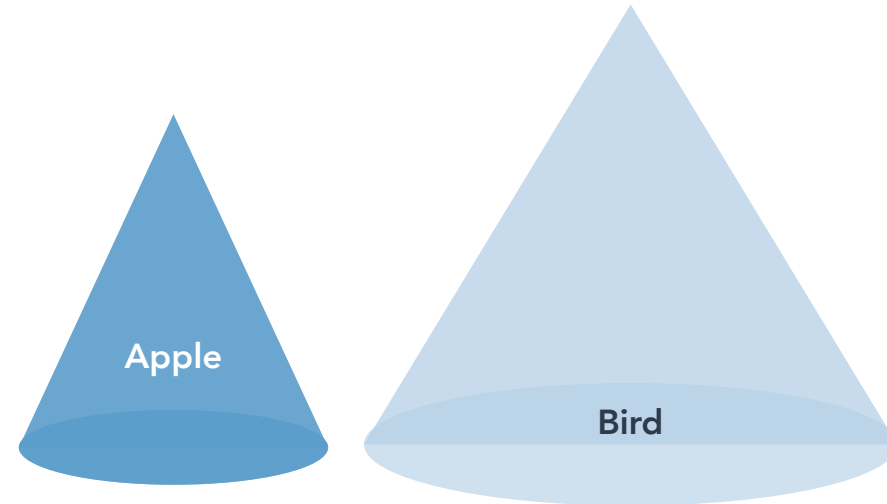
Inside the cone



"swallow" is a "bird"

inside the cone → is-a relations

Outside the cone



"apple" is not a "bird"

outside → it is not

Training: Learning the Cones

Learn the cones from known **is-a** and **not-is-a** pairs examples

Energy = how far a word pokes outside its cone. Zero if the word sits safely inside.

Pull friends in

True pairs: pull the word
inside the cone

Push strangers out

False pairs: push the word
outside the cone

The Oracle's Final Decision

Three steps to accept an "is-a" axiom

1

Measure

How far the word sits
inside or outside the
cone

2

Score

Turn distance into a
confidence between 0~1:
inside > 0.5
outside < 0.5

3

Decide

A word has many senses:
try all pairings, keep the
best score

Accept the "is-a" axiom when the **best score is above 0.5**.

Experiments: Data & Oracle Training

Building the WordNet dataset

Source WordNet — extract hypernym–hyponym edges, clean, add negatives, split

Test set only pairs that exist in Japanese (our target language)

Basic edges

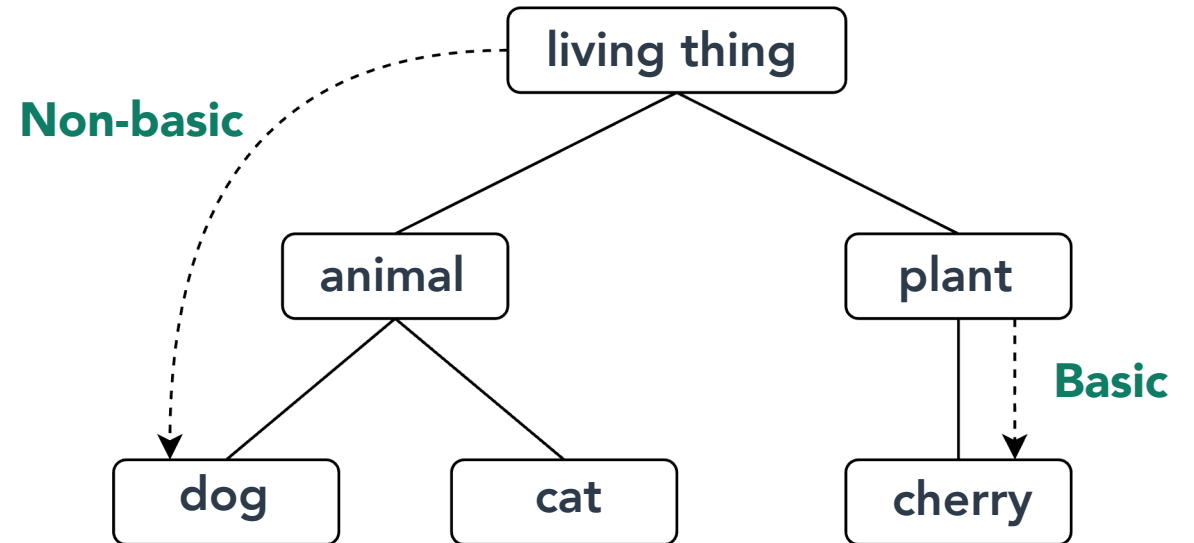
Direct parent–child (plant → cherry).
Keep all to hold the hierarchy.

Non-basic edges

Longer ancestor–descendant (living thing → dog).
Vary the amount to test the effect.

Evaluate by F1

Order Embedding · Poincaré · Hyperbolic Cones



Classifier results by F1 score

Reported in the paper (Ganea+ 2018, Table 1):

Model	Dim = 5					Dim = 10				
	0%	10%	25%	50%	90%	0%	10%	25%	50%	90%
Order Emb	34.4	70.2	75.9	81.7	–	43.0	69.7	79.4	84.1	–
Poincaré	29.4	70.2	78.2	83.6	–	28.9	71.4	82.0	85.3	–
Hyp Cones	29.2	80.1	86.0	92.8	–	32.2	85.9	91.0	94.4	–

Our Japanese data:

Model	Dim = 5					Dim = 10				
	0%	10%	25%	50%	90%	0%	10%	25%	50%	90%
Order Emb	38.0	72.7	78.3	83.3	87.4	46.0	72.4	81.4	87.0	87.0
Poincaré	28.6	55.2	73.8	79.4	82.4	30.2	68.0	82.3	84.5	86.3
Hyp Cones	29.9	73.2	75.1	92.0	93.3	30.9	76.9	87.9	92.8	93.9

The Oracle in Action

Best setting (10 dims, 90% non-basic edges) reaches **93%** on the binary test

"vehicle" $\rightarrow$ "fire engine" = **0.52**
Entailment

"bird" $\rightarrow$ "apple" = **0.14**
Non-entailment

Before integrating Oracle:

Premise "Taro eats an apple"
Hypothesis "Taro eats fruit"

$\rightarrow$ **Unknown**

Premise 1 "Taro eats an apple"
Premise 2 "An apple is fruit"
Hypothesis "Taro eats fruit"

$\rightarrow$ **Yes**

Inside the Formal Proof

Built into the prover, the Oracle supplies the missing step **automatically**

$$\begin{array}{c}
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash s_0 : \left[\begin{array}{l} u_3 : \left[\begin{array}{l} x_3 : \text{entity} \\ x_4 : \text{entity} \\ \text{apple}(x_4, x_3) \end{array} \right] \\ x_5 : \text{entity} \\ \text{eat}(x_5, \text{Taro}, \pi_1(u_3)) \end{array} \right]} \quad (\text{Var}) \\
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash \pi_1(\pi_2(\pi_2(\pi_1(s_0)))) : \text{apple}(\pi_1(\pi_2(\pi_1(s_0))), \pi_1(\pi_1(s_0)))} \quad (\Sigma E) \\
 \frac{}{s_0 : \left[\begin{array}{l} u_0 : \left[\begin{array}{l} x_0 : \text{entity} \\ x_1 : \text{entity} \\ \text{apple}(x_1, x_0) \end{array} \right] \\ x_2 : \text{entity} \\ \text{eat}(x_2, \text{Taro}, \pi_1(u_0)) \end{array} \right] \vdash \text{oracle}(\pi_1(\pi_2(\pi_2(\pi_1(s_0)))) : \text{fruit}(\pi_1(\pi_2(\pi_1(s_0))), \pi_1(\pi_1(s_0)))} \quad (\text{Con})
 \end{array}$$

Premise 1 "Taro eats an apple"
Hypothesis "Taro eats fruit"

→ **Yes**

Impact on the full system

Dataset: 20 upward-entailing monotonicity cases from MED, translated into Japanese

Before Oracle

rows = predictions · columns = gold

	Yes	Unk
Yes	0	0
Unk	10	10

all predicted Unknown

After Oracle

	Yes	Unk
Yes	8	1
Unk	2	9

17 / 20 correct

Accuracy

50% → 85%

Yes recall

0% → 80%

Yes precision

≈89%

High precision $\Rightarrow$ the Oracle adds knowledge safely, **without poisoning the proofs**

Error analysis 1

Cause 1: Out-of-vocabulary (OOV) words

What happens

An OOV word appears in the pair



Not in WordNet's Japanese entries



Oracle defaults to score = 0.0

Typical OOV words

Proper nouns

Place names

Personal names

Rare words

DIAGNOSIS

A coverage problem — not a flaw in the geometry

FIX

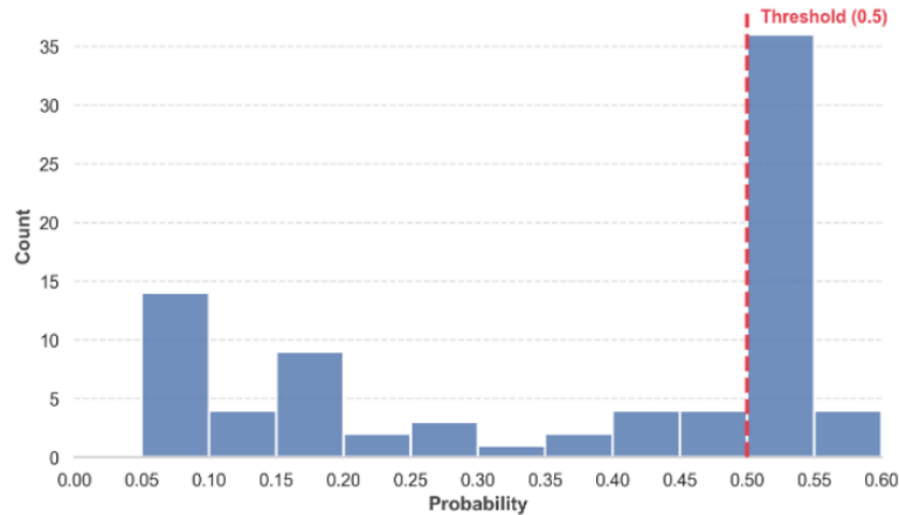
Train on higher-coverage resources (e.g. KWJA)

Error analysis 2

Cause 2: The type of relation

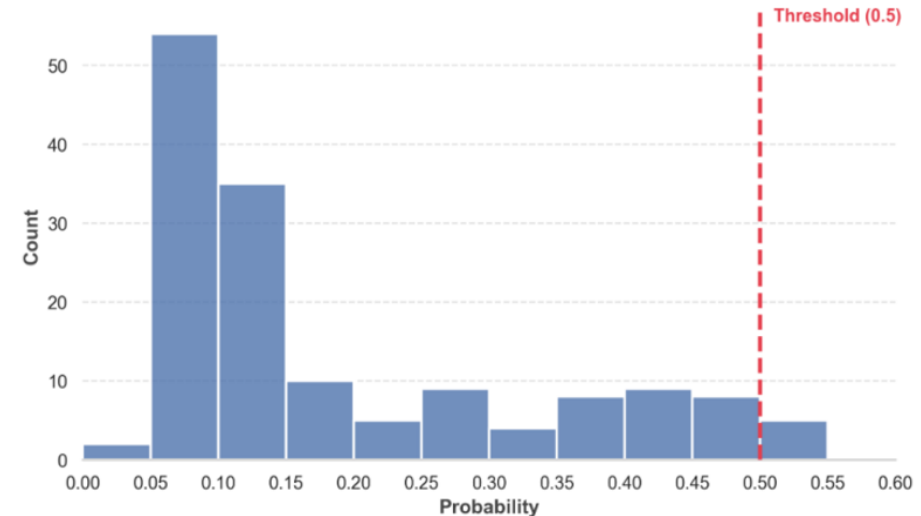
Scored 280 hypernym–hyponym pairs from MED, and inspected the misclassified ones, leaving out the OOV cases.

Positive pairs (N = 100)



Accuracy : 62.12%

Negative pairs (N = 180)



Accuracy : 96.64%

Great at saying “No” — but it misses some true positives

Error analysis 2

Why positives are missed — and why that is by design

Positive (acc. 62.12%)

Most missed pairs are “**part-of**” relations, not “is-a”.

These matter for NLI, but they are out of the cone model’s scope by design.

Negative (acc. 96.64%)

~97% accuracy means almost no wrong axioms — so we keep the **threshold at 0.5**.
The model stays silent rather than guessing wrong.

Conclusion

A neuro-symbolic NLI system:

the prover keeps its guarantees while Oracle supplies lexical knowledge dynamically.

WHAT WE DID

- Embedded knowledge with Hyperbolic Entailment Cones
- Integrated Oracle into the proving pipeline
- Implemented the Oracle in Haskell

KEY FINDING

- Oracle injects knowledge safely — high precision, with no hallucinated axioms.

FUTURE WORK

- Scale to larger resources (e.g. KWJA) for broader vocabulary
- Handle multi-argument predicates
- Explore other hyperbolic approaches for part-of and further relations

Acknowledgements

Part of this work was supported by JST CREST JPMJCR2565 and JSPS KAKENHI JP23H03452.

References

1. Bekki, D. and Mineshima, K.: Context-Passing and Underspecification in Dependent Type Semantics, *Studies in Linguistics and Philosophy*, pp. 11–41 (2017).
2. Bekki, D., Tanaka, R. and Takahashi, Y.: Integrating Deep Neural Networks with Dependent Type Semantics, *LACompLing2021* (2021).
3. Bekki, D., Tanaka, R. and Takahashi, Y.: Learning Knowledge with Neural DTS, *NALOMA III*, pp. 17–25 (2022).
4. Daido, H. and Bekki, D.: Development of an automated theorem prover for the fragment of DTS, *LENLS17* (2020).
5. Bond, F., Baldwin, T. and Uchimoto, K.: Japanese SemCor, *GWC-2012*, pp. 56–63 (2012).
6. Ganea, O., Bécigneul, G. and Hofmann, T.: Hyperbolic Entailment Cones for Learning Hierarchical Embeddings, *ICML 35*, pp. 1646–1655 (2018).
7. Martin-Löf, P.: *Intuitionistic Type Theory*, Vol. 9, Bibliopolis (1984).
8. Nickel, M. and Kiela, D.: Poincaré Embeddings, *NIPS 2017*, pp. 6338–6347 (2017).
9. Socher, R., Chen, D., Manning, C. D. and Ng, A.: Reasoning with neural tensor networks for knowledge base completion, *NIPS 2013*, pp. 926–934 (2013).
10. Steedman, M.: *Surface Structure and Interpretation*, MIT Press (1996).
11. Steedman, M.: *The Syntactic Process*, MIT Press (2000).
12. Tomita, A., Matsubara, M., Daido, H. and Bekki, D.: NLI with CCG Parser and ATP for DTS, *BriGap-2*, pp. 1–7 (2025).
13. Vendrov, I., Kiros, R., Fidler, S. and Urtasun, R.: Order-Embeddings of Images and Language, *ICLR 2016* (2016).
14. Daido, H.: Introduction of equality types into an ATP for a DTS fragment, Master's thesis, Ochanomizu University (2022).
15. Iinuma, M. and Bekki, D.: Implementation of the learning algorithm of the Neural DTS NLI system, *Proc. 31st Annual Meeting of the ANLP*, pp. 488–492 (2025).

Appendix A : cone geometry

Each concept u is embedded in the Poincaré ball $\mathbb{D}^n$, excluding an open ball $\mathcal{B}^n(O, \varepsilon)$ ($\varepsilon > 0$) around the origin. A learnable $K > 0$ parameterizes the cone's half-aperture:

$$\psi(u) = \sin^{-1}(K \cdot (1 - \|u\|^2) / \|u\|), \quad K \leq \varepsilon / (1 - \varepsilon^2)$$

Entailment condition: $\Xi(u, v) \leq \psi(u)$, where the geodesic angle is

$$\Xi(u, v) := \pi - \angle O_{uv} = \cos^{-1}(N / D)$$

$$N = \langle u, v \rangle (1 + \|u\|^2) - \|u\|^2 (1 + \|v\|^2)$$

$$D = \|u\| \|u - v\| \sqrt{(1 + \|u\|^2)\|v\|^2 - 2\langle u, v \rangle}$$

$\|\cdot\|, \langle \cdot, \cdot \rangle$: Euclidean norm and inner product.

Appendix B : training objective & decision rule

Energy (how far v sticks out of u 's cone; 0 if inside):

$$E(u, v) = \max(0, \Xi(u, v) - \psi(u))$$

Loss ($\mathbb{P}$: positive pairs, $\mathbb{N}$: negative pairs, margin γ):

$$\mathcal{L} = \sum_{(u,v) \in \mathbb{P}} E(u, v) + \sum_{(u',v') \in \mathbb{N}} \max(0, \gamma - E(u', v'))$$

Entailment probability between vectors:

$$\delta(v_1, v_2) = 1 / (1 + e^{\Xi(v_1, v_2) - \psi(v_1)})$$

Decision rule ($S(w)$: set of sense vectors of word w):

$$\text{oracle}(w_1, w_2) = \max_{v_1 \in S(w_1), v_2 \in S(w_2)} \delta(v_1, v_2), \quad \text{valid axiom iff } \text{oracle}(w_1, w_2) > 0.5$$

Appendix C : framework for Dynamic Axiom Addition

During backward inference, wani decides on the fly whether to call the hyperbolic Oracle to add a lexical axiom:

- 1** A predicate subgoal appears
Backward inference produces a goal like $\text{dog}(s, x)$ — a predicate applied to a term. This triggers the classifier.
- 2** Find a related premise
Search the premises for the same term x under a different predicate, e.g. $\text{puppy}(s, x)$.
- 3** Ask the Oracle
Pass puppy (hyponym candidate) and dog (hypernym candidate) to the classifier.
- 4** Add the axiom if confident
If the probability > 0.5 , accept $\text{puppy} \sqsubseteq \text{dog}$ as a valid axiom and resolve the subgoal.